

ROOM ALLOCATION SYSTEM

ExamRoomandSeat AllocationManagementSystem

Project Abstract • Object-Oriented Programming

Title Page

Project title: Room Allocation System – Exam Room and Seat Allocation Management System

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Introduction

- **Brief introduction to the project** – The Exam Room Allocation Management System is an object-oriented software application designed to automate the process of assigning exam rooms and seats to students during examinations. The system models real-world entities such as Student, Room, Seat, Exam and Admin as classes to build a clean, modular and extensible solution.
- **Purpose of developing the system** – To replace slow and error-prone manual seating work with a centralized, automatic system that generates fair, conflict-free and malpractice-resistant seating arrangements and reduces the workload of the exam cell.
- **Technology used** – Java (Object-Oriented Programming), JDBC for database connectivity, MySQL / PostgreSQL as the backend database, developed using Eclipse / IntelliJ IDEA / NetBeans IDE.

Problem Statement

- **Existing problem** – Manually allocating rooms and seating for examinations is tedious and difficult to manage, especially with multiple exams, large student strength and limited rooms.
- **Limitations of the existing / manual system** – Time-consuming and error-prone; risk of overcrowding, seat clashes and double allocation; unfair proximity of students of the same class / section or roll-number range which raises malpractice concerns; no centralized records and difficulty in preparing seating charts.
- **Why a computerized system is required** – A computerized system is required to automate allocation quickly and accurately, enforce capacity and fairness constraints, avoid conflicts, maintain centralized student, room and exam data, and generate room-wise student lists and seating charts instantly.

Objectives

- **Main objectives of the project** – To develop a centralized exam room and seat allocation system that accepts, processes and displays student and room data to automatically generate seating arrangements.
- **What the system aims to achieve** – Fair distribution of students across rooms and seats based on class / department mixing, roll-number sequencing and room capacity limits; conflict-free scheduling across multiple exams; validation of seating capacity; and generation of seating charts and room-wise lists through an admin interface.
- **Expected benefits** – Reduced manual workload, minimized human error, prevention of overcrowding and clashes, malpractice-resistant seating, quick reports, and practical demonstration of OOP concepts – encapsulation, inheritance, polymorphism and abstraction – in Java.

Proposed System

- **Overview of the proposed solution** – A Java-based OOP application in which each Room has a defined seating capacity and layout, and the allocation logic distributes students across rooms and seats as per defined constraints. Different room types (e.g., regular classroom vs. hall) are modelled using inheritance with shared base behaviour, keeping the design extensible.
- **Main features** – Automatic seat allocation; room capacity validation; conflict-free scheduling for multiple exams; class / department mixing with roll-number-based or random-shuffling strategies (using polymorphism); generation of seating charts / room-wise student lists; and an admin interface to input exam details and available rooms.
- **How the proposed system addresses the problem** – By enforcing constraints during allocation, the system ensures adjacent students are not from the same course / section, prevents allocation beyond capacity, eliminates seat clashes, and replaces manual registers with a centralized database for fast, accurate and repeatable allocation.

System Requirements

Hardware requirements:

- Processor – Intel Core i3 or above
- RAM – 4 GB minimum (8 GB recommended)
- Storage – 500 MB free disk space
- Standard display, keyboard and mouse
- Printer (optional, for printing seating charts)

Software requirements:

- IDE: Eclipse / IntelliJ IDEA / NetBeans
- Programming language: Java
- Database: MySQL / PostgreSQL
- JDBC or other database connectivity technology
- Operating System: Windows / Linux (any OS supporting Java)

System Architecture / Design

User (Admin) → Java Application → JDBC → Database

- **User (Admin / Exam Cell)** interacts with the Java Application to enter student details, exam details and available rooms, and to view or print the generated allocation.
- **Java Application** contains the OOP classes (Student, Room, Seat, Exam, Admin) and the allocation logic. It validates capacity, applies constraints such as class mixing and roll-number sequencing, and generates the seating plan.
- **JDBC** acts as the bridge between the Java Application and the Database, handling storage and retrieval of records securely and efficiently.
- **Database** stores master data (students, rooms, seats, exams) and allocation results persistently, and returns data on demand for seating charts and room-wise student lists.

End of Abstract